Example 4.7

If the selling price of an LED TV is equal to $\frac{5}{4}$ of its cost price, then find the gain / profit percentage.

Solution:



Let the C.P of the LED TV be ₹x.

$$\therefore \text{ S.P} = \frac{5}{4}x$$

$$\text{Profit} = \text{S.P} - \text{C.P} = \frac{5}{4}x - x = \frac{x}{4}$$

$$\text{Profit \%} = \left(\frac{Profit}{C.P} \times 100\right)\%$$

$$\therefore \text{ Profit \%} = \left(\frac{x/4}{x} \times 100\right)\% = \left(\frac{1}{4} \times 100\right)\% = 25\%$$

Example 4.8

The cost price of 16 boxes of strawberries is equal to the selling price of 20 boxes of strawberries. Find the gain or loss percentage.

Solution:

Let the C.P of one strawberry box be ₹ x .

C.P of 20 strawberry boxes = $20x$ and

S.P of 20 strawberry boxes = C.P of 16 strawberry boxes = $16x$ (given)



Here, S.P < C.P, hence there is a loss.

$$\text{Loss} = \text{C.P} - \text{S.P} = 20x - 16x = 4x$$

$$\text{Loss \%} = \left(\frac{Loss}{C.P} \times 100\right)\%$$

$$= \left(\frac{4x}{20x} \times 100\right)\%$$

$$\therefore \text{ Loss \%} = 20 \%$$



If the cost price of x articles = selling price of y articles, then profit % $= \left(\frac{x-y}{y} \times 100\right)\%$ .If the answer is negative, then it is treated as loss. Check and verify this for Example 4.8.

Example 4.9

By selling a bicycle for ₹4275, a shopkeeper loses 5%. For how much should he sell it to have a profit of 5%?

04_8th_Maths_Life Mathematics_Chap 4.indd 128 20/11/2022 13:12:49